

Assignment 4 - Customer, Product & Order Analysis

Revenue is Price x Quantity; the supplied tables are complete, duplicate-free, and have valid foreign keys.

Q1. Load all three CSV files.

Answer:

```
customers = pd.read_csv("customers.csv")
products = pd.read_csv("products.csv")
orders = pd.read_csv("orders.csv")
```

Q2. Display the dimensions of each DataFrame.

Answer:

customers: **3,000 x 5**; products: **100 x 4**; orders: **15,000 x 5**.

Q3. Identify the columns in each dataset.

Answer:

Customers: Customer_ID, Customer_Name, City, Age, Gender. Products: Product_ID, Product_Name, Category, Price. Orders: Order_ID, Customer_ID, Product_ID, Quantity, Order_Date.

Q4. Identify the common columns between datasets.

Answer:

Customer_ID links customers to orders. Product_ID links products to orders. There is no direct customer-products key.

Q5. Determine how the three datasets are related.

Answer:

One customer can have many orders; one product can appear in many orders. Each order row references one customer and one product.

Q6. Check each DataFrame for missing values.

Answer:

All three tables have **zero missing values**.

Q7. Check for duplicate records.

Answer:

Customers: 0; Products: 0; Orders: 0 exact duplicate rows.

Q8. Check validity of Customer_ID and Product_ID.

Answer:

Every order references a valid customer and product: **0 orphan customer IDs and 0 orphan product IDs**.

Q9. Convert Order_Date into datetime.

Answer:

```
orders["Order_Date"] = pd.to_datetime(orders["Order_Date"], errors="coerce")
All 15,000 dates parse.
```

Q10. Check whether every order corresponds to a valid customer.

Answer:

orders.Customer_ID.isin(customers.Customer_ID).all() is True; unmatched count = **0**.

Q11. Check whether every order corresponds to a valid product.

Answer:

orders.Product_ID.isin(products.Product_ID).all() is True; unmatched count = **0**.

Q12. Merge orders with customers.

Answer:

```
orders_customers = orders.merge(customers, on="Customer_ID", how="left", validate="many_to_one")
```

Q13. Merge the resulting DataFrame with products.

Answer:

```
final_df = orders_customers.merge(products, on="Product_ID", how="left", validate="many_to_one")
Result: <b>15,000 rows</b>.
```

Q14. Create a final consolidated DataFrame with order, customer, and product details.

Answer:

```
final_df["Order_Date"] = pd.to_datetime(final_df["Order_Date"])
final_df = final_df[["Order_ID", "Customer_ID", "Customer_Name", "City", "Age", "Gender", "Product_ID", "Product_Name", "Category"]]
```

Q15. Create Order_Value = Price x Quantity.

Answer:

```
final_df["Order_Value"] = final_df["Price"] * final_df["Quantity"]
```

Q16. Calculate total revenue.

Answer:

Sum of Order_Value across 15,000 orders: **\$2,233,171,871.30**.

Q17. Calculate average order value.

Answer:

Total Order_Value / 15,000 orders = **\$148,878.12**.

Q18. Find total revenue by product category.

Answer:

Beauty \$296,691,801.75; Books \$305,056,331.77; Clothing \$426,148,894.21; Electronics \$331,292,623.06; Home Appliances \$331,310,958.87; Sports \$542,671,261.64.

Q19. Find total revenue by city.

Answer:

Bangalore \$263,705,876.95; Chennai \$251,968,512.26; Coimbatore \$283,122,456.36; Delhi \$265,454,070.70; Hyderabad \$293,263,918.87; Kolkata \$315,549,989.08; Mumbai \$280,356,550.10; Pune \$279,750,496.98.

Q20. Find the top 10 customers by spending.

Answer:

Customer_ID (name), total spend: C00019 (Customer_19) \$3,176,078.87; C02897 (Customer_2897) \$2,528,868.53; C01915 (Customer_1915) \$2,476,932.74; C02709 (Customer_2709) \$2,400,905.35; C00187 (Customer_187) \$2,394,078.09; C00685 (Customer_685) \$2,352,464.63; C01297 (Customer_1297) \$2,345,815.39; C00259 (Customer_259) \$2,330,070.43; C00921 (Customer_921) \$2,257,528.60; C00271 (Customer_271) \$2,226,186.64.

Q21. Find the top 10 products by revenue.

Answer:

Product_74 \$47,258,003.68; Product_88 \$47,213,343.92; Product_3 \$47,075,130.00; Product_46 \$45,000,771.20; Product_52 \$44,990,386.56; Product_73 \$44,897,635.86; Product_68 \$44,696,626.32; Product_38 \$43,658,393.09; Product_54 \$40,946,977.89; Product_50 \$40,756,127.88.

Q22. Find the most frequently purchased product.

Answer:

By total units ordered, **Product_15** is most purchased, with 600 units.

Q23. Find the city with the highest number of customers.

Answer:

Kolkata, with **412 customers**.

Q24. Find the city generating the highest revenue.

Answer:

Kolkata, \$315,549,989.08.

Q25. Divide customers into age groups: 18-30, 31-45, 46-60, 61+.

Answer:

Use inclusive age bands 18-30, 31-45, 46-60, and 61+. Among customers with orders: 18-30: 736; 31-45: 849; 46-60: 873; 61+: 518. There are 24 registered customers with no orders.

Q26. Calculate revenue generated by each age group.

Answer:

18-30: \$558,627,811.38; 31-45: \$622,352,370.11; 46-60: \$659,923,687.41; 61+: \$392,268,002.40.

Assignment 4 - Customer/Order Analysis (Q27-Q35)

Q27. Compare spending between male and female customers.

Answer: Total customer spend: Female \$1,078,971,684.20; Male \$1,154,200,187.10. Mean lifetime spend/customer: Female \$738,010.73; Male \$762,351.51.

Q28. Identify high-value customers.

Answer: Define high-value at or above the 90th percentile of lifetime spend across **all 3,000 registered customers** (including zero-spend customers). Cutoff: **\$1,336,722.25**; 300 customers meet/exceed it.

Q29. Determine the percentage of customers who placed more than one order.

Answer: 2,881 of all 3,000 registered customers placed multiple orders: **96.03%**. Among customers with at least one order, it is 96.81% (2,881/2,976).

Q30. Create a customer spending ranking.

Answer: Group Order_Value by customer and sort descending. Top: C00019 (Customer_19), \$3,176,078.87.

Q31. Identify the top 10% of customers based on total spending.

Answer: Select the 300 highest-spending customers out of 3,000 registered customers (10%).

Q32. Find the most profitable category for each city.

Answer: Product costs are not supplied, so profit is not computable. Using revenue as a proxy, Sports is the highest-revenue category in all eight cities.

Q33. Find the most popular product in each category.

Answer: By total units sold: Sports Product_96; Clothing Product_15; Books Product_63; Electronics Product_54; Beauty Product_10; Home Appliances Product_17.

Q34. Calculate monthly revenue.

Answer: 2025-01 \$53,361,220.93; 2025-02 \$52,963,079.79; 2025-03 \$56,233,288.12; 2025-04 \$50,923,849.46; 2025-05 \$57,301,343.61; 2025-06 \$56,460,489.60; 2025-07 \$54,157,148.69; 2025-08 \$57,144,165.05; 2025-09 \$53,352,193.95; 2025-10 \$56,630,127.31; 2025-11 \$59,276,825.16; 2025-12 \$59,630,660.46; 2026-01 \$56,260,151.26; 2026-02 \$47,689,351.93; 2026-03 \$55,497,703.15; 2026-04 \$51,306,881.32; 2026-05 \$51,823,753.71; 2026-06 \$51,618,514.74; 2026-07 \$54,128,777.26; 2026-08 \$57,438,462.80; 2026-09 \$55,460,907.01; 2026-10 \$57,171,177.74; 2026-11 \$56,875,351.32; 2026-12 \$52,435,333.48; 2027-01 \$56,072,466.01; 2027-02 \$48,032,412.73; 2027-03 \$51,740,476.06; 2027-04 \$51,673,329.49; 2027-05 \$55,972,559.14; 2027-06 \$55,791,489.47; 2027-07 \$57,720,544.86; 2027-08 \$55,334,425.53; 2027-09 \$53,134,726.53; 2027-10 \$53,646,489.78; 2027-11 \$53,680,865.96; 2027-12 \$55,292,934.52; 2028-01 \$52,205,366.58; 2028-02 \$48,474,671.11; 2028-03 \$55,381,632.73; 2028-04 \$51,636,778.77; 2028-05 \$57,204,319.10; 2028-06 \$5,035,625.08.

Q35. Identify the month with the highest revenue.

Answer: December 2025, \$59,630,660.46.